

QUESTION PAPER

Name of the Examination: CAT 2 (WIN 2022-2023)

Course Code: CSE2011

Course Title: Cyber Security & Digital Forensics

Slot: 2

Date of Exam: 27/03/2023 (FN)

Duration: 90 min

Total Marks: 50

(AI)

1. What are the key technical components of an email spoofing mitigation strategy, and how should they be prioritized? [10]
2. What are the limitations of ChatGPT in cybersecurity, and how can they be addressed to improve its effectiveness? [10]
3. What are the key compliance and regulatory requirements that apply to DDoS mitigation strategies for corporate private networks, and how can they be met? [15]
4. How can memory analysis be used to detect and analyze advanced persistent threats (APTs) and other sophisticated attacks that may be designed to evade traditional security controls? Justify your answer. [15]

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1, 2, 3	1, 2	1, 3	-	-	10
Q2	3	3, 4	1, 2, 3	-	-	10
Q3	3	1, 2	1, 3	-	-	15
Q4	4	1, 2	1, 3	-	-	15

QUESTION PAPER

Name of the Examination: CAT 2 (WIN 2022-2023)

Course Code: CSE2011

Course Title: Cyber Security & Digital Forensics

Slot: 3

Date of Exam: 03/04/2023 (Fm)
(G1)

Duration: 90 min

Total Marks: 50

1. What are the potential business impacts of a successful email spoofing attack, and how can these impacts be mitigated? [10]
2. What are the potential use cases for ChatGPT in cybersecurity, and how can it be used to enhance threat intelligence and analysis? [10]
3. What are the key indicators that a DDoS attack is in progress on a corporate private network, and how can they be detected? Justify your answer with a proper mitigation plan. [15]
4. How can network traffic analysis be used in combination with other forensic techniques, such as endpoint analysis or memory analysis, to provide a more complete picture of an attack?. What are the potential risks and impacts of malicious network traffic, and how can organizations assess and prioritize these risks? [15]

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1,2,3	1, 2	1, 3	-	-	10
Q2	1,2,3	3, 4	1, 2, 3	-	-	10
Q3	3	1, 2	1, 3	-	-	15
Q4	4	1, 2	1, 3	-	-	15



QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT 2

Course Code: CSE2011

Course Title: Cyber Security & Digital Forensics

Set number: 4

Date of Exam: 29/03/2023 (AN)

Duration: 90 Min

Total Marks: 50

(C2)

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Can you explain what a role is in context with RBAC? Why are roles important? Is it possible to assign multiple roles to a single user? If yes, then how? Can you give some examples of real-world applications that use RBAC? **(15M)**

Q2 What if my personal email account, bank account, or other accounts were compromised? How to report a Phishing or suspicious email? Justify with an Example? **(15M)**

Q3 What is the purpose of analysing and monitoring Network Traffic? What to look for in network traffic analysis and monitoring Solutions? **(10M)**

Q4 When attempting to obtain someone's information, what types of social engineering tactics are usually utilized less than simply hacking someone's software? **(10M)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	1	1	1	1	15
Q2	3	1	1	1	1	15
Q3	4	2	2,3	2	2	10
Q4	4	3	4,5	2	2	10

QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT 2

Course Code: CSE2011

Course Title: CyberSecurity & Digital Forensics

Set number: 5

Date of Exam: 22/03/2023 (AN)

Duration: 90 Min

Total Marks: 50 (A2)

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1 What happens if there's an error while assigning permissions to users? How can we prevent privilege escalation attacks when using RBAC? How can we make sure that only authorized users have access to sensitive data? Justify with an Example? **(15M)**

Q2 What happens if you click on a phishing email link or attachment? Why do I need to watch out for phishing emails? What should you do if you're unsure whether an email is real? **(15M)**

Q4 What are three common methods of social Engineering? What are common warning signs of social Engineering? Describe your own words? **(10M)**

Q5 How can we preserve volatile evidence at the crime scene? Many say MD5 has been compromised and should not be used for an integrity check in digital forensics. What is your opinion? If yes what is your solution? **(10M)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	1	1	1	1	15
Q2	3	1	1	1	1	15
Q3	4	2	2,3	2	2	10
Q4	4	2	4,5	2	2	10